WRYM · The Worm Arena

A deterministic, WOIM-style grid-worm benchmark for language models — certified by an exhaustive solver, archived as a static site, open to any model via MCP

Whitepaper v3.1 · 2026-11-05 · 31 certified levels · 31 archived runs · ladder peak 9.0/10 NIGHTMARE

1 · Abstract

We introduce WRYM, a benchmark in which language models play a deterministic grid-worm puzzle game in the design tradition of **WOIM** (Dogspawn, 2026): eat every apple, then reach the heart. Around that loop the game wraps the classic toolbox of the hardest public puzzlers — sokoban-style crate pushes, weighted plates and gates, keys and padlocks, one-way valves, crumbling bridges, diggable dirt, frictionless ice, lethal spikes — and one signature mechanic: the worm can **cut itself in half**, leaving inert remains that carry weight and block slides. Models observe the board as a text projection and emit one action per turn. Every level is certified by an exhaustive BFS solver (solvability, optimal line, state count); every recorded run is re-executed by a deterministic checker and signed. A seven-dimension, evidence-based difficulty index measures each level from its certified optimal line — depth, search width, irreversibility, simulation, threat, choreography and budget — and refuses to inflate what a level does not contain. The ladder currently spans 31 levels across six tiers (mean difficulty 5.3), topping out at The Rubicon, certified 9.0 / NIGHTMARE, engineered from primary research on Snakebird, Stephen's Sausage Roll, the Sokoban PSPACE-completeness construction, and WOIM itself. The entire archive — runs, replays, model thoughts, certificates — renders from the repository as a static GitHub Pages site with zero backend, and a browser Level Forge lets anyone contribute levels via pull request, certified by the same engine that runs CI.

TL;DR. Deterministic worm puzzler, text-only observations, one action per turn. Exhaustive solver certifies every level and its par; deterministic checker signs every run; seven-dimension difficulty index scores the ladder honestly (peak: **9.0 NIGHTMARE**). Any model plugs in via OpenAI-style adapters or MCP. Runs execute locally; GitHub Pages renders everything — leaderboard, level atlas, certified replays with the model's mind alongside the board. Build levels in the browser Forge and submit them by pull request.

2 · Why a worm game

Grid puzzle games are a clean instrument for measuring planning in language models: fully observable, compactly serialisable state; deterministic rules without dice or timers; difficulty that is structural rather than perceptual. The worm genre adds the ingredient most benchmarks lack — **the agent's own body is part of the state**. Eating apples lengthens the worm (sometimes into a trap); the split action converts half the body into inert terrain that can hold a plate down or stop a slide. Complexity theory takes this family seriously: item-collecting Snake is PSPACE-complete (De Biasi, 2018) and Sokoban is PSPACE-complete (Culberson, 1997), so the puzzles this engine generates live in the same neighbourhood as the canonical hard planning problems — while remaining small enough to certify exhaustively.

3 · The game

A level is a rectangular grid over a nineteen-character vocabulary. The rules, in brief: invalid moves consume no turn and return a reason (free information, exactly as the real game teaches); apples grow the worm by one segment; the heart is sealed until every apple is eaten; crates push sokoban-style onto floor, plates, bridges and valves; plates are weighted by crates, remains, or any worm segment; gates open while every plate is weighted; one-way valves may only be entered along their arrow; bridges crumble to rubble the instant they are vacated; entering ice starts a slide that continues until the head lands on grippy ground or is stopped — sliding

into spikes is fatal; keys are picked up by crawling over them and consumed by padlocks; the split keeps the front half (with the head) and drops the back half as remains. Each level carries its own turn limit.

glyph	element	semantics
#	soil wall	impassable bedrock
.	tunnel	carved open floor
@ / o	worm head / body	exactly one head; grows with apples
*	apple	eat all to unseal the heart
X	heart goal	the win tile
$	walnut crate	pushable, parks on plates
%	floor button	weighted by crate / remains / worm
D	golden gate	open while every button is pressed
~	spikes	instant death, also mid-slide
+	soft dirt	diggable, permanent
k / L	key / padlock	inventory + consumption
> < ^ v	one-way valves	enter along the arrow only
=	crumble bridge	cross once, then rubble
_	frictionless ice	slide until stopped

4 · The benchmark contract

An agent observes a text serialisation of the full game state plus the rule book, and emits one action per turn. Runs are scored on levels solved, moves against par, deaths, and wasted turns; pairwise Elo across models is computed per level and folded into a leaderboard. Resilience telemetry (retries, rate waits, degradation, tokens, cost) is archived per call. The local half is the repository: bun scripts/run-benchmark.ts executes any configured model or the exhaustive **wyrm-bfs** baseline and writes JSON artifacts. The remote half is GitHub: CI re-certifies the ladder on every push and pull request, and Pages renders the archive — leaderboard, level atlas, and a replay viewer that shows the board and the model's mind side by side, certified frame by frame. A Model Context Protocol server (mcp/llm-bridge.mjs) exposes the game as tools, so any MCP-capable client can play without custom integration code.

5 · The difficulty index (v3)

Difficulty is measured, not asserted. The index replays the certified optimal solution through the real engine and records evidence: plan length; BFS states explored; every permanent world mutation (digs, unlocks, valve lock-ins, burned bridges, splits); every forced ice glide; and, at every state along the line, each legal alternative probed for fatality. Seven dimensions saturate smoothly to 100 — **depth**, **search** (log scale), **irreversibility**, **simulation**, **threat**, **choreography** (sub-goal ordering load), **budget** (turn-limit pressure) — and the composite blends their weighted mean with the strongest dimension, mapping to six tiers from TUTORIAL to NIGHTMARE. The index refuses to inflate: a level with no ice can never score on simulation; no spikes, never on threat. Reaching 9.0 requires extremity in every dimension simultaneously — which is why one level in thirty-one achieves it.

6 · Tier V — engineering the extreme

The five levels of the Fifth Circle were designed against primary sources rather than intuition. From **Snakebird** we took one-wrong-press unwinnability and fruit-order length contracts; from **Stephen's Sausage Roll**, chains of irreversible acts; from the **Sokoban PSPACE proof**, corridor registers — one-way commitment structures; from **nuworm**'s design essays, action at a distance; and from **WOIM** itself, the cut-yourself-in-half economy where flesh is a resource. The certified optimal lines then exhibited behaviours we did not explicitly design: in Absolute Zero and The Rubicon the worm splits mid-route so that its own inert remains become the stop-block

the next ice slide requires — body-as-infrastructure emerging from the rules rather than the layout. The full dossier, with citations, ships as docs/DESIGN_RESEARCH.md.

Top of the certified ladder:

level	tier	score	par	states	signature
The Rubicon	NIGHTMARE	9.0	38/40	23,992	all seven ≥ 65
Absolute Zero	ABYSS	8.2	20/21	4,287	
Glass Foundry	ABYSS	8.1	18/20	261,073	
The Reliquary	GAUNTLET	7.9	30/32	44,069	
Six Seals	GAUNTLET	7.8	27/29	121,335	
Cathedral Of Lead	GAUNTLET	7.7	62/65	107,545	
Serpents Choir	GAUNTLET	7.5	22/24	261,270	
Wyrm Genocide	GAUNTLET	7.3	45/54	38,315	

7 · The fully-static archive

The deliverable is the repository itself. Local runs write results/ as JSON; CI certifies and the export script compiles the site's data; GitHub Pages serves it with no backend, no database and no build-time secrets. The Level Forge page ships the same engine, validator and difficulty index — compiled from the identical source CI runs — so a level that certifies in a contributor's browser is, by construction, the level CI will certify. Submitted levels arrive as pull requests to community-levels/, are certified by the verify job, and upon merge join the public ladder automatically. Reproducibility is not a property of the artefact; it is the user interface.

8 · Snapshot

certified levels	31
tier spread	TUTORIAL 5 · CORE 8 · ADVANCED 8 · GAUNTLET 7 · ABYSS 2 · NIGHTMARE 1
archived runs (current export)	31
configured models	2 (incl. any-model-via-MCP)
ladder peak	9.0 / NIGHTMARE — 31-the-rubicon
mean difficulty	5.3 / 10

9 · References

Culberson, J. (1997). Sokoban is PSPACE-complete. · De Biasi, M. (2018). The complexity of snake and undirected NCL variants. Theoretical Computer Science. · Hearn, R. & Demaine, E. (2002). PSPACE-completeness of sliding-block puzzles. arXiv:cs/0205005. · Dogspawn (2026). WOIM. store.steampowered.com/app/4272020. · Noumenon Games (2015). Snakebird. · increpare (2016). Stephen's Sausage Roll. · noa (2023). Journey to nuworm. cubestudio.itch.io/nuworm. · Icely Puzzles (2026). Would You Love Me If I Was a Snake? store.steampowered.com/app/4284670.